LSTM Feature Extraction

```
import tensorflow as tf
from tensorflow.keras.datasets import imdb
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, LSTM, Dense
import matplotlib.pyplot as plt
vocab_size=10000
(x_train, y_train), (x_test, y_test) = imdb.load_data(num_words = vocab_size)
max_length = 200
X_train = pad_sequences(x_train, maxlen = max_length)
X_test = pad_sequences(x_test, maxlen = max_length)
model = Sequential()
model.add (Embedding(input_dim = vocab_size, output_dim=
                input_length = max_length))
model.add (LSTM(64)
model.add (Embedding (input_dim=vocab_size, output_dim=64,
                input_length = max_length))
model.add (LSTM (64))
model.add (Dense (1, activation='sigmoid'))
model.compile (optimizer = 'adam'
               loss = 'binary_crossentropy',
               metrics = ['accuracy'])
history = model.fit (x_train, y_train, epochs=5,
               batch_size= 64, validation_split = 0.2)
loss, accuracy = model.evaluate(x_test, y_test)
print ("Test Accuracy :", accuracy)
plt.plot (history.history['accuracy'], label = "Training
                Accuracy")
plt.plot (history.history['val_accuracy'], label='validation
                Accuracy")
plt.legend()
plt.xlabel ("Epoch")
plt.ylabel ("Accuracy")
plt.title ("LSTM Model Accuracy")
plt.show()
```

OUTPUT

Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/imdb.npz

Epoch 1/5

/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:97: UserWarning: Argument 'input_length' is deprecated. Just remove it.

Epoch: 2/5

313/313 accuracy: 0.9044 loss: 0.2513 val_accuracy: 0.8702 val_loss: 0.3042

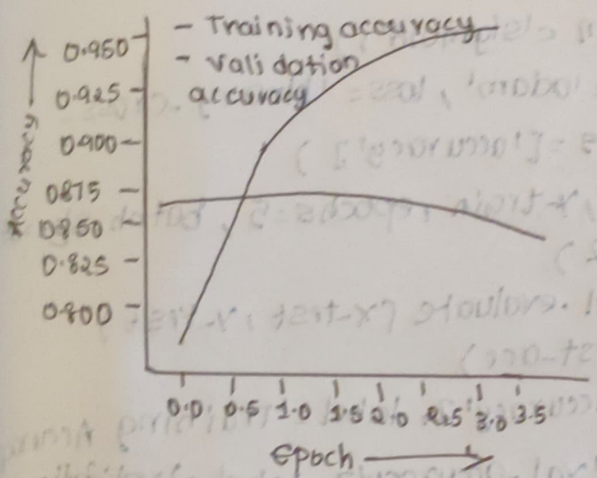
Epoch: 3/5

313/313 accuracy: 0.9367 loss: 0.1721 val_accuracy: 0.8658 val_loss: 0.3211

Epoch: 4/5

312/313 accuracy: 0.9569 loss: 0.1238 val_accuracy: 0.8662 val_loss: 0.3865

Epoch: 5/5

313/313 accuracy: 0.9617 loss: 0.1082 val_accuracy: 0.8492 val_loss: 0.3928

782/782 accuracy: 0.8515 loss: 0.3974